

Lecture 12

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conditioning v. imp. for midsem

Conditioning on an event A ($A \in \mathcal{F}$)

$$P_{X|A}(x) = \frac{P(\{X=x\} \cap A)}{P(A)}$$

$$\sum_x P_{X|A}(x) = 1$$

consistency

$$E[X|A] = \sum_x x \cdot P_{X|A}(x)$$

$$E[g(x)|A] = \sum_x g(x) \cdot P_{X|A}(x)$$

Conditioning X with disjoint partitions $\{A_i\}$ of Ω

$$P_X(x) = \sum_i P(A_i) \cdot P_{X|A_i}(x)$$

* Multiplying both sides with x and taking summation,

$$\sum_x x \cdot P_X(x) = \sum_x x \cdot \sum_i P(A_i) \cdot P_{X|A_i}(x)$$

$$E[X] = \sum_i P(A_i) \cdot E[X|A_i]$$

Conditioning X on an event $X \in A$ ($A \in \mathcal{F}'$)

$$X \in A = \{\omega \in \Omega : X(\omega) \in A\}$$

$$P\{X \in A\} = \sum_{x \in A} P_X(x)$$

$$P_{X|A}(x) = \frac{P(\{X=x\} \cap \{X \in A\})}{P(X \in A)}$$

If $x \notin A$, we have $P_{X|A}(x) = 0$

Otherwise, when $x \in A$, we have $P_{X|A}(x) = \frac{P_X(x)}{P(X \in A)}$

Revisiting geometric random variable:

$$P_N(k) = (1-p)^{k-1} \cdot p$$

Suppose event $A: N > n$

$$\therefore P(A) = (1-p)^n$$

$$P_{N|A}(k) = \begin{cases} (1-p)^{k-n-1} \cdot p, & \text{if } k > n \\ 0, & \text{otherwise} \end{cases}$$

Conditioning X on a random variable Y

We want prob. that $X=x$ given $Y=y$.

$$P_{X|Y=y}(x) = \frac{P_{XY}(x, y)}{P_Y(y)} = P_{X|Y}(x|y)$$

$$P_{XY}(x, y) = P_Y(y) \cdot P_{X|Y}(x|y)$$

$$= P_X(x) \cdot P_{Y|X}(y|x)$$

$$\sum_x P_{X|Y}(x|y) = \frac{\sum_x P_{XY}(x|y)}{P_Y(y)} = \frac{P_Y(y)}{P_Y(y)} = 1$$

* If X and Y are independent:

$$P_{XY}(x, y) = P_X(x) \cdot P_Y(y)$$

$$P_{X|Y}(x) = P_X(x)$$

$$P_{Y|X}(y) = P_Y(y)$$

Independent $\Rightarrow$ uncorrelated ($\text{Cov}(X, Y) = 0$)

Uncorrelated $\nRightarrow$ independent

We know $P_{XY}(x, y) = P_{X|Y}(x|y) \cdot P_Y(y)$

$\Rightarrow$ Summing on both sides over y ,

$$P_X(x) = \sum_y P_Y(y) \cdot P_{X|Y}(x|y)$$

Similarly, $P_{XY}(x, y) = P_{X|Y}(x|y) \cdot P_X(x)$

$\Rightarrow$ Summing on both sides over x ,

$$P_Y(y) = \sum_x P_X(x) \cdot P_{Y|X}(x|y)$$